CLASS 10 MATHEMATICS • CHAPTER 5

Arithmetic Progressions

Exhaustive point-by-point presentation covering patterns, general terms, sums of n terms, and word problems directly from the textbook.

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Patterns in Nature & Daily Life

🌱 Patterns Around Us

- Sunflower petals, honeycombs, pineapples, and pine cones follow striking regular patterns.
- Reena's salary: ₹8000 with annual increment of ₹500 (8000, 8500, 9000...).
- Ladder rungs decreasing uniformly by 2 cm (45, 43, 41...).

🧮 Savings & Fibonacci Sequence

- Shakila's savings box: ₹100 starting with ₹50 yearly increments.
- Fibonacci rabbit breeding sequence: 1, 1, 2, 3, 5, 8...
- We study patterns where succeeding terms are obtained by adding a fixed number.

Section 5.1 Live Check

Q1. Can you name two natural objects that exhibit mathematical patterns or spirals?

Q2. What is the pattern rule in Reena's salary progression: 8000, 8500, 9000?

Q3. How is the Fibonacci sequence generated using the rabbit pair example?

Definition, Common Difference & General Form

Arithmetic Progression (AP) Definition

An AP is a list of numbers in which each term is obtained by adding a **fixed number** (common difference d) to the preceding term except the first term (a).

General Form: a, a + d, a + 2d, a + 3d, ...
Common Difference: d = a_(k+1) - a_k

Section 5.2 Live Check

Q1. What is an arithmetic progression, and what name is given to the fixed added number?

Q2. Can the common difference of an AP be negative or zero? Give examples.

Q3. How do you check if a given list of numbers forms an AP?

Deriving the General Term formula

General Term Formula

If the first term is a and common difference is d, the nth term (or general term) a_n is given by:

$$a_n = a + (n - 1)d$$

Key Concepts

- $a_1 = a$, $a_2 = a + d$, $a_3 = a + 2d$, ..., $a_n = a + (n - 1)d$.
- In a finite AP with m terms, a_m represents the last term (l).

Section 5.3 Live Check

Q1. What is the formula for finding the nth term of an AP?

Q2. Why do we multiply the common difference by (n − 1) instead of n when finding a_n?

Q3. What alternative symbol is frequently used for the last term of a finite AP?

Sum Formula & Gauss's Technique

Young Gauss's Trick

- At age 10, Carl Friedrich Gauss summed 1 to 100 by writing forward and reverse rows.
- Sum = (n/2)[2a + (n - 1)d]
- Useful for adding dozens or thousands of terms instantly!

√× Key Sum Formulas

$$S_n = (n/2)[2a + (n - 1)d]$$
$$S_n = (n/2)(a + l)$$
$$a_n = S_n - S_{(n-1)}$$

Section 5.4 Live Check

Q1. What are the two primary formulas for finding the sum of the first n terms of an AP?

Q2. When is the formula $S_n = (n/2)(a + l)$ most convenient to use?

Q3. How can you find the nth term of an AP if you are given the formula for the sum of its first n terms (S_n)?

Worked Exercise Highlights

Problem 1: Sum to n terms (Ex 5.3 Q3(i))

- Given $a = 5$, $d = 3$, $a_n = 50$. Find n and S_n.
- $a_n = a + (n - 1)d \Rightarrow 50 = 5 + (n - 1)3 \Rightarrow n = 16$.
- $S_{16} = (16/2)(5 + 50) = 8 \times 55 = \mathbf{440}$.

Problem 2: 31st Term of AP (Ex 5.2 Q7)

- 11th term is 38 ($a + 10d = 38$).
- 16th term is 73 ($a + 15d = 73$).
- Solving gives $d = 7$, $a = -32$.
- $\mathbf{a_{31} = -32 + 30(7) = 178}$.

Exercise Highlights Live Check

Q1. How do we form simultaneous linear equations when given two specific terms of an AP?

Q2. Once you find 'a' and 'd', what is the fastest way to compute any required term like the 31st term?

Q3. Why is checking your intermediate step for n (ensuring it is a positive integer) crucial in word problems?

Chapter Recap & Checklist

AP Foundations

- **Definition:** Constant common difference $d = a_{(k+1)} - a_k$.
- **General Form:** $a, a + d, a + 2d, a + 3d...$

Core Formulas

- **nth Term:** $a_n = a + (n - 1)d$.
- **Sum of n Terms:** $S_n = (n/2)[2a + (n - 1)d]$ or $(n/2)(a + l)$.

THANK YOU

"It is as if humanity has been set a puzzle, the solution of which we are working out, stone by stone." — W.W. Sawyer

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